

shortened. To this observer, it seems that the small ball isn't pulling away from the big ball very fast.

Now here's what's interesting about all this. The outcome shown in figure al/2 was supposed to be the only one possible, the only one that satisfied both conservation of energy and conservation of momentum. So how can the *different* result shown in figure am be possible? The answer is that relativistically, momentum must not equal mv . The old, familiar definition is only an approximation that's valid at low speeds. If we observe the behavior of the small ball in figure am, it looks as though it somehow had some extra inertia. It's as though a football player tried to knock another player down without realizing that the other guy had a three-hundred-pound bag full of lead shot hidden under his uniform — he just doesn't seem to react to the collision as much as he should. This extra inertia is described⁶ by redefining momentum as

$$p = m\gamma v.$$

At very low velocities, γ is close to 1, and the result is very nearly mv , as demanded by the correspondence principle. But at very high velocities, γ gets very big — the small ball in figure am has a γ of 2.1, and therefore has more than double the inertia that we would expect nonrelativistically.

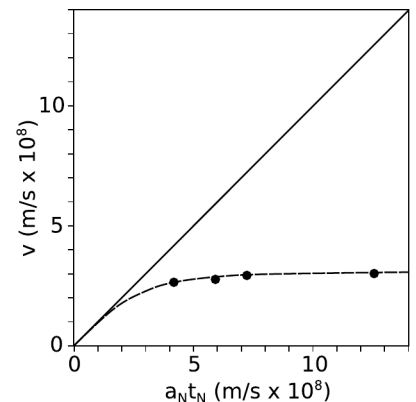
This also explains the answer to another paradox often posed by beginners at relativity. Suppose you keep on applying a steady force to an object that's already moving at $0.9999c$. Why doesn't it just keep on speeding up past c ? The answer is that force is the rate of change of momentum. At $0.9999c$, an object already has a γ of 71, and therefore has already sucked up 71 times the momentum you'd expect at that speed. As its velocity gets closer and closer to c , its γ approaches infinity. To move at c , it would need an infinite momentum, which could only be caused by an infinite force.

Push as hard as you like . . .

example 7

We don't have to depend on our imaginations to see what would happen if we kept on applying a force to an object indefinitely and tried to accelerate it past c . A nice experiment of this type was done by Bertozzi in 1964. In this experiment, electrons were accelerated by an electric field E through a distance ℓ_1 . Applying Newton's laws gives Newtonian predictions a_N for the acceleration and t_N for the time required.⁷

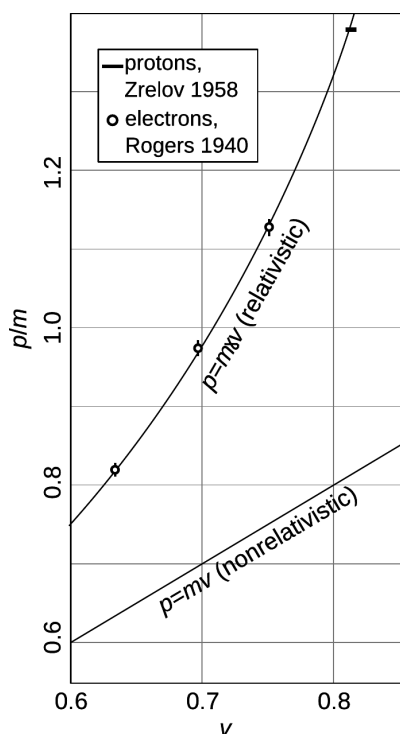
The electrons were then allowed to fly down a pipe for a further distance $\ell_2 = 8.4$ m without being acted on by any force. The time of flight t_2 for this second distance was used to find the final velocity $v = \ell_2/t_2$ to which they had actually been accelerated.



an / Example 7.

⁶See p. 805 for a proof.

⁷Newton's second law gives $a_N = F/m = eE/m$. The constant-acceleration equation $\Delta x = (1/2)at^2$ then gives $t_N = \sqrt{2m\ell_1/eE}$.



26.10 / Two early high-precision tests of the relativistic equation $p = \gamma mv$ for momentum. Graphing p/m rather than p allows the data for electrons and protons to be placed on the same graph. Natural units are used, so that the horizontal axis is the velocity in units of c , and the vertical axis is the unitless quantity p/mc . The very small error bars for the data point from Zrelov are represented by the height of the black rectangle.

Figure 26.9 shows the results.⁸ According to Newton, an acceleration a_N acting for a time t_N should produce a final velocity $a_N t_N$. The solid line in the graph shows the prediction of Newton's laws, which is that a constant force exerted steadily over time will produce a velocity that rises linearly and without limit.

The experimental data, shown as black dots, clearly tell a different story. The velocity never goes above a certain maximum value, which we identify as c . The dashed line shows the predictions of special relativity, which are in good agreement with the experimental results.

Figure 26.10 shows experimental data confirming the relativistic equation for momentum.

Equivalence of mass and energy

Now we're ready to see why mass and energy must be equivalent as claimed in the famous $E = mc^2$. So far we've only considered collisions in which none of the kinetic energy is converted into any other form of energy, such as heat or sound. Let's consider what happens if a blob of putty moving at velocity v hits another blob that is initially at rest, sticking to it. The nonrelativistic result is that to obey conservation of momentum the two blobs must fly off together at $v/2$. Half of the initial kinetic energy has been converted to heat.⁹

Relativistically, however, an interesting thing happens. A hot object has more momentum than a cold object! This is because the relativistically correct expression for momentum is γmv , and the more rapidly moving atoms in the hot object have higher values of γ . In our collision, the final combined blob must therefore be moving a little more slowly than the expected $v/2$, since otherwise the final momentum would have been a little greater than the initial momentum. To an observer who believes in conservation of momentum and knows only about the overall motion of the objects and not about their heat content, the low velocity after the collision would seem to be the result of a magical change in the mass, as if the mass of two combined, hot blobs of putty was more than the sum of their individual masses.

We know that the masses of all the atoms in the blobs must be the same as they always were. The change is due to the change in γ with heating, not to a change in mass. The heat energy, however, seems to be acting as if it was equivalent to some extra mass. If the quantity of heat is E , then it turns out that the extra mass m is

⁸To make the low-energy portion of the graph legible, Bertozzi's highest-energy data point is omitted.

⁹A double-mass object moving at half the speed does not have the same kinetic energy. Kinetic energy depends on the square of the velocity, so cutting the velocity in half reduces the energy by a factor of $1/4$, which, multiplied by the doubled mass, makes $1/2$ the original energy.